MILITARY SPECIFICATION

PROJECTILE, 8-INCH, HE, RA, M650
AFT WARHEAD SECTION AND STRIPPABLE COATING ASSY.

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for the 8-Inch Aft Warhead Section and Strippable Coating Assy., for the use in Projectile, 8-Inch, HE, RA, M650.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this Military Specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	-Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coating

MILITARY

MIL-A-2550	-Ammunition, General Specification for
MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-S-50783	-Steel Alloy - Special Purpose for Ammunition Components (HF-1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-141	-Paint, Varnish, Lacquer and Related Materials Method of Inspection and Testing

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	-Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1167	-Ammunition Data Card
MIL-STD-1168	-Lot Numbering of Ammunition

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKAGING DRAWINGS

9288037	-Aft Warhead Section and Strippable Coating Assembly
9280132	-Pallet for Aft Warhead, M650

INSPECTION EQUIPMENT DRAWINGS

9280266	-Envelope Drawing UT Equipment
9246488	-Ultrasonic Standard, Radial and Longitudinal, M650
9246489	-Ultrasonic Standard, Circumferential, M650
9287968	-Warhaed M650, Hydrostatic Test Assembly
9287967	-Hydrostatic Test Fixture, M650

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E8	-Tension Testing of Metallic Materials
E18	-Test for Rockwell Hardness of Metallic Materials
A711	-Carbon Steel Blooms, Billets and Slabs for Forging

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

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ANSI B46.1 -Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

PANGBORN ROTO BLAST STEEL SHOT AND GRIT CATALOG

(Copies may be obtained from Pangborn Division of the Carborundum Co., Hagerstown, MD 21740).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawing 9288037 and associated drawings and with all requirements specified in applicable specifications and standards.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9288037 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Heat treatment. Heat treatment or stress relief shall be uniform throughout a lot of warheads. No warhead shall be reheat treated without prior Government approval.

3.4 Slow bend precracked charpy. The nominal fracture toughness shall be greater than 25 KSI $\sqrt{in}$ (informational only).

3.5 Warhead material defects.

3.5.1 Warhead body. The warhead body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams, and other metal defects.

3.5.2 Warhead cavity. In addition to the requirements of 3.5.1, the warhead cavity shall be free from scale, fins, burns, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.5.3 Visual standards. Visual standards for evaluation of metal defects will be established for each contract after the first month's production. However, establishment of visual standards shall not waive the requirements for compliance with 3.5.1 and 3.5.2 (see 6.9).

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3.6 Hydrostatic test. The warhead shall withstand an internal hydrostatic pressure in accordance with requirements specified on Dwg. 9287943 without evidence of leakage, cracks, ruptures or permanent deformation.

3.7 Ultrasonic inspection. The warheads shall be free of any discontinuities equal to or in excess of the inspection criteria within their respective zones. The inspection criteria are notches and holes in standards (see Dwg. 9246488 and 9246489) which simulate rejectable cracks and piping flaws. (That is, all projectiles shall have an echo signal that is less than the signal obtained from the standards).

3.8 Interior cavity finish.

3.8.1 The interior cavity after painting shall be free from bare spots, grease, dirt, or other foreign matter.

3.8.2 There shall be no blisters or pool of paint in the bottom of the cavity prior to palletizing.

3.9 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI-B46.1 and/or Pangborn Shot and Grit catalog shall be used as a basis for surface roughness determination.

3.10 Bonding assembly pull test. The warhead insulation shall withstand a pull test of 100 pounds, minimum.

3.11 Insulation. The insulation shall be free from cracks, seams, porosity, cavities and other defects.

3.12 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.13 Proving ground. There shall be no break-up or separation of metal parts in the gun bore or in flight.

3.14 Workmanship. The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and on the following:

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3.14.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.14.2 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with Drawing 9287943.

3.14.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of specification MIL-A-2550.

3.14.4 Burr. No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.14.5 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawings	Quantity
Aft Warhead Section and Strippable Coating Assembly	9288037	25
Warhead (Unpainted Metal Parts)	9287943	25
Warhead Insulation	9287965	10 Consecutively produced from each cavity of each mold.
Aft Warhead Section and Strippable Coating Assembly (for Ballistic Testing, see 4.3.2.1)	9288037	20
Disk	9288038	10

All assemblies and components produced by the contractor or furnished by a supplier shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A First Article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.2.1 Ballistic testing. Twenty complete assemblies packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government installation designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1.).

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48322B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Aft Warhead Section and Insulation Assy				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Aft Warhead Section and Strippable Coating Assy. (Dwg. 9288037)				
	Examination for Defects	25		3.2	4.4.2.9/4.4.2.7/4.4.2.5
	Bonding Assy. Pull Test	(a)		3.10	4.5.12
	Salt Spray Test	2		3.2	4.5.10
	Proving Ground Test	20		3.13	4.5.13
	Warhead (Prior to Painting) Metal Parts (Dwg. 9287943)				
	Examination for Defects	25		3.2	4.4.2.2/4.4.2.3/4.4.2.4/ 4.4.2.6
	Visual Inspection	(b)		3.5	4.4.2.4
	Hydrostatic Test	(b)		3.6	4.5.7
	Ultrasonic Test (d)	(b)		3.7	4.5.8
	Mechanical Properties Test	2		3.2	4.5.1/4.5.2
	Base Cavity Radius	(c)		3.2	4.5.5
	Interior Surface Finish	(c)		3.9	4.5.6
	Warhead Insulation (Dwg. 9287965)				
	Examination for Defects	10/mold		3.2	4.4.2.1
	Material Defects	(e)		3.11	4.4.2.1
	Disk (Dwg. 9288038)				
	Examination for Defects	10		3.2	4.4.2.8

NOTES: (a) Use 25 assy. utilized for examination for defects.
(b) Use 25 assy. utilized for examination for defects.
(c) Use 2 assy. utilized for mechanical properties test.
(d) Excluding base thread area.
(e) Use 1/mold selected from the components utilized for examination for defects.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, for the first three lots produced, heat treatment inspection lots shall not exceed any of the following: One mill-heat of steel, eight hours production or 500 components. In any event, heat treatment inspection lots shall be restricted to components heat treated in the same furnace at the same temperature and for the same length of time. Test samples shall be selected after final heat treatment. Where heat treatment is performed at the same time in a batch-type furnace, additional identification control must be maintained which directly relates to that portion of the inspection lot processed as a batch. The method of control must meet with approval of the Government. Components in heat treatment inspection lots shall remain grouped together for identifying purposes until mechanical property tests and interior surface and internal radius examinations have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots.

4.2.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48322B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Insulation, Warhead		9287965
			NEXT HIGHER ASSEMBLY 9288019

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Large outside diameter	(a)		3.2	Gage
102	Outside diameter of step	(a)		3.2	Gage
103	Small outside diameter on taper	(a)		3.2	Gage
104	True position of outside diameter of step	(a)		3.2	Gage
105	True position of outside diameter on taper	(a)		3.2	Gage
106	Thickness of outer edge	(a)		3.2	Gage
107	Thickness of step	(a)		3.2	Gage
108	Overall thickness	(a)		3.2	Gage
109	Surface finish of flat side glazed or improper	(a)		3.2	Visual
110	Flatness of bottom surface	(a)		3.2	Gage
111	Material defects	(b)		3.5	Visual
Minor					
201	Radius at bottom of taper missing or improper	(a)		3.2	Visual
202	Radius at large outside diameter	(a)		3.2	Visual
203	Evidence of poor workmanship	(a)		3.14	Visual

NOTES:
(a) See paragraph 4.4.3.6.
(b) See paragraph 4.4.3.7.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Warhead, Prior to Base Threading, Knurling and Painting				9287943
					NEXT HIGHER ASSEMBLY 9288019
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Hydrostatic test		100%	3.6	4.5.7
2	Ultrasonic test		100%	3.7	4.5.8
3	Material Defects	a		3.5	4.5.6/Visual
Major					
101	Mechanical properties (hardness)		100%	3.2	4.5.1
102	Mechanical properties (tension)	b		3.2	4.5.2
103	Slow bend precrack charpy test	c		3.4	4.5.3
104	Steel quality	d		3.1	4.5.4
105	Radius of base cavity	e		3.2	4.5.5

NOTES: (a) See paragraph 4.4.3.5
(b) See paragraph 4.4.3.1.2
(c) See paragraph 4.4.3.2
d. See paragraph 4.4.3.3
e. See paragraph 4.4.3.4

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48322B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Warhead, Prior to Knurling				9287943
					NEXT HIGHER ASSEMBLY 9288019
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Pitch diameter of base thread		100%	3.2	Gage
102	Major diameter of base thread, minimum		100%	3.2	Gage
103	Runout of pitch diameter of base thread with bourrelet (two locations) including seating of gage on shoulder		0.40%	3.2	Gage
104	Length of base thread, including thread undercut		0.40%	3.2	Gage
105	Length of thread undercut		0.40%	3.2	Gage
Minor					
201	Distance from front face to rear of step adjacent to undercut		0.65%	3.2	Gage
202	Distance from front face to rear shoulder		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48322B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.4	Warhead, Prior to Painting				9287943
					NEXT HIGHER ASSEMBLY
					9288019
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Hydrostatic test mark missing		100%	3.2	Visual
2	Material Defects		100%	3.5	4.5.9/Visual
3	Ultrasonic Test Mark Missing		100%	3.2	Visual
Major					
101	Pitch diameter of nose thread		0.40%	3.2	Gage
102	Diameter of nose thread undercut		0.40%	3.2	Gage
103	Runout of pitch diameter of nose thread with bourrelet (in two locations), including seating of gage on shoulder		0.40%	3.2	Gage
104	Runout of smallest basic diameter at nose end with bourrelet		0.40%	3.2	Gage
105	Runout of largest basic diameter of ogive with bourrelet		0.40%	3.2	Gage
106	Diameter of bourrelet, minimum		0.40%	3.2	Gage
107	Thickness of base, min.		0.40%	3.2	Gage
108	Wall thickness at basic gage distance nearest nose (360°)		0.40%	3.2	Gage
109	Wall thickness at basic gage distance nearest base (360°)		0.40%	3.2	Gage
110	Wall thickness at middle basic gage distance (360°)		0.40%	3.2	Gage
111	Diameter of base thread undercut		0.40%	3.2	Gage

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	CONTINUATION SHEET				9287943
					NEXT HIGHER ASSEMBLY 9288019
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Runout of base thread undercut with bourrelet		0.40%	3.2	Gage
113	Surface finish in base thread undercut		0.40%	3.2	Gage
114	Diameter of step adjacent to nose thread undercut		0.40%	3.2	Gage
115	Smallest internal diameter at open end of cavity		0.40%	3.2	Gage
116	Knurl missing or improper		100%	3.2	Visual
Minor					
201	Diameter of recess in base		0.65%	3.2	Gage
202	Depth of recess in base		0.65%	3.2	Gage
203	Thickness of base, max.		0.65%	3.2	Gage
204	Distance from front face to basic diameter at nose		0.65%	3.2	Gage
205	Distance from front face to largest basic diameter on ogive		0.65%	3.2	Gage
206	Length of nose thread, including undercut		0.65%	3.2	Gage
207	Length of nose thread undercut		0.65%	3.2	Gage
208	Chamfer on nose thread missing or improper		0.65%	3.2	Visual
209	Chamfer on base thread missing or improper		0.65%	3.2	Visual
210	Outside diameter of datum on chamfer on nose		0.65%	3.2	Gage
211	Runout of outside diameter of datum on chamfer on nose		0.65%	3.2	Gage
212	Evidence of poor workmanship		1.0%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Warhead, After Painting		9287943
			NEXT HIGHER ASSEMBLY 9288019

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Visual inspection (after phosphate coating but prior to paint)		100%	3.5	Visual/4.5.9
2	Pool of paint in bottom of cavity or blisters		100%	3.8	Visual/4.5.9
Major					
101	Diameter of bourrelet, max. (after paint)		100%	3.2	Gage
102	Interior protective coating missing, improper or inadequate, including bare spot		100%	3.8	Visual
103	Grease, dirt or other foreign matter in cavity		100%	3.8	Visual
104	Salt spray test	(a)		3.2	Visual/4.5.10
105	Paint thickness	(b)		3.2	Gage/4.5.11
Minor					
201	Painted surface with total exposed or damaged area in excess of 1/4 sq. inch		0.65%	3.2	Visual
202	Marking missing, misleading, or inidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.14	Visual

NOTES:
(a) See paragraph 4.4.3.9
(b) See paragraph 4.4.3.10

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48322B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Aft Warhead Section Prior to Assembly of Insulation				9288019
					NEXT HIGHER ASSEMBLY
					9288037
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Warhead base improperly finished		0.40%	3.2	Visual
102	Warhead base not clean or dry		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48322B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Aft Warhead Section and Insulation Assembly		9288019
			NEXT HIGHER ASSEMBLY
			9288037

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Insulation missing or loose		0.40%	3.2	Visual/Manual
102	Insulation or adhesive above flush		0.40%	3.2	Gage
103	Bonding assembly pull test		100%	3.10	4.5.12
Minor					
201	Evidence of poor workmanship		1.0%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48322B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Disk				9288038
					NEXT HIGHER ASSEMBLY
					9288037
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major	None defined.				
Minor					
201	Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.00%	3.14	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Aft Warhead Section and Strippable Coating Assembly		9288037
			NEXT HIGHER ASSEMBLY 9288022

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Proving ground test	(a)		3.13	4.5.13
Major	None defined.				
Minor					
201	Tape missing or improperly applied		0.65%	3.2	Visual
202	Disk missing or improper		0.65%	3.2	Visual
203	Strippable coating missing or improper		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.00%	3.14	Visual

NOTES:
(a) See paragraph 4.4.3.8

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48322B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Pallet				9288037
					NEXT HIGHER ASSEMBLY
					9288022
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major	None defined.				
Minor					
201	Pallet damaged (unserviceable)		0.65%	3.2	Visual
202	Improper packing		0.65%	3.2	Visual
203	Strapping missing, loose or broken		0.65%	3.2	Visual
204	Marking missing, incorrect or illegible		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.14	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Hardness. Each final heat treated warhead from a heat treatment lot shall be serialized and subjected to the hardness test. Hardness readings shall be recorded for each serialized warhead and the average hardness reading determined. The average hardness reading shall be used in selected the tension samples of paragraph 4.4.3.1.2.

4.4.3.1.2 Tension. Two final heat treated warheads from a heat treat lot shall be selected for the tension test. The warheads selected shall be those with the highest and lowest average hardness readings from the hardness test of paragraph 4.4.3.1.1. From each warhead, two specimens shall be tested in accordance with paragraph 4.5.2.

4.4.3.1.3 Retest. In the event of rejection, new samples of higher or lower average hardness or both, as applicable, may be selected and tested in accordance with paragraph 4.4.3.1.2 in order to establish an acceptable hardness range. When the hardness range is established, all warheads with hardness values outside the range shall be rejected or subjected to re-temper if approved by the Government. An acceptable hardness range is one that provides an acceptable yield and elongation.

4.4.3.2 Slow bend precracked charpy test.

4.4.3.2.1 Initial production testing. Beginning with the initial lot, the warheads selected for testing in accordance with 4.4.3.1 for high hardness shall have eight specimens taken and tested as follows:

Base Section. Four specimens shall be taken as detailed in Figure 1. Two (2) specimens shall be tested at minus sixty-five degrees fahrenheit (-65°F) and two (2) specimens at seventy degrees fahrenheit (70°F). The notches shall be machined on the outside surface of the specimen.

Sidewall. Four specimens shall be selected longitudinally from between stations X and Y of Drawing 9287943. Two (2) samples each shall be tested at minus sixty-five degrees fahrenheit (-65°F) and seventy degrees fahrenheit (70°F). The notches shall be machined on the outside surface of the specimen.

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4.4.3.2.2 After 100 acceptable heat treat lots have been produced, the sample size shall be reduced to four samples per heat treat. Only the minus sixty-five degree fahrenheit (-65°F) test shall be performed.

4.4.3.3 Steel quality. The macroetch test shall be performed on the top end of the top and on the bottom end of the bottom cuts from the first, middle and last ingate. Material failing to comply with the requirement specified on Drawing (Dwg.) 9287943 shall be rejected. The contractor shall certify that the steel meets the macroetch requirement of Drawing 9287943 and MIL-S-50783.

4.4.3.4 Radius of base cavity. The warheads used in 4.4.3.1.2 shall be gaged for proper radius when the warheads are cut for tension specimens. If any warhead fails to comply, the heat treated inspection lot shall be rejected.

4.4.3.5 Interior surface. The warheads used in paragraph 4.4.3.1.2 shall be examined for interior surface discontinuities after sectioning. Failure to meet the requirement of paragraph 3.5 shall be cause for rejection of the heat treat inspection lot.

4.4.3.6 Molded components. The first ten (10) warhead insulations consecutively produced from each cavity of each mold shall be inspected for all drawing dimensions. In addition, ten (10) consecutive sample pieces shall be selected and inspected after a maximum of 5000 pieces have been produced. If any part fails to meet all drawing requirements, the production quantity represented by the sample shall be rejected. Deviations or suspected deviations in product shall be examined to determine if corrective action is required.

4.4.3.7 Void examination. The first insulation from each mold shall be cut in half and examined for the requirement in 3.11. If the piece is defective, each piece that follows shall be cut in half and examined until an acceptable part is produced. Additionally, one piece from each 200 produced from each mold shall also be cut in half and examined in accordance with 3.5.11. If any piece is defective, that quantity represented by the sample shall be rejected.

4.4.3.8 Proving ground. Security - Testing shall be performed as specified in 4.5.13.

4.4.3.8.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with all of the applicable requirements specified, 20 sample warheads

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shall be randomly selected for this test. The 20 sample warheads shall be observed for security of parts when tested in accordance with 4.5.13. If any of the sample warheads fail to comply with the requirements for security of parts, the lot shall be rejected.

4.4.3.8.2 After three consecutive lots have complied with 4.4.3.8.1, 10 sample warheads shall be randomly selected from each lot for this test. The 10 sample warheads shall be observed for security of parts. If any of the sample warheads fail to comply with the requirement for security of parts, the lot shall be rejected.

4.4.3.9 Salt spray. Salt spray samples shall be selected alternately, one during the first 30 minutes of production and one randomly at different times of the day. Failure to comply with the requirements of the applicable drawing shall be cause for taking the action recommended by the applicable specification.

4.4.3.10 Protective coating - thickness. Two samples shall be selected at random to represent each four-hour period, or less, of continuous paint production. The sample warheads shall be inspected using the equipment in accordance with 4.5.11. The sample warheads shall be inspected for paint coating thickness at a minimum of four separate locations on each warhead. Any warhead failing to meet the applicable requirement shall be classed defective and every warhead painted since the last acceptable sample shall be rejected.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Mechanical properties - hardness. The hardness test shall be conducted in accordance with ASTM Method E18. One reading shall be taken at location X and one at location Y as indicated on Dwg. 9287943.

4.5.2 Mechanical properties - tension testing.

Base section. The base section tensile specimen shall be taken transversely, one inch from the centerline. Failure to meet the yield strength or elongation requirements of Dwg. 9287943 shall require a retest from the remaining portion of the

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base section. Results of the retest specimen and the first failing specimen shall be averaged to determine compliance with the applicable drawing requirement. Failure of the average results to meet the drawing requirement shall be cause for rejection of the heat treated inspection lot. In addition, failure of either individual test specimen to meet the 115,000 psi min. yield strength and 3% min. elongation shall be cause for rejection of the heat treatment inspection lot represented by the sample.

Side wall. One specimen shall be taken from between station X and Y. Failure of any specimen to comply with the applicable drawing requirement shall be cause for rejection of the heat treated inspection lot.

Testing shall be conducted in accordance with The American Society for Testing and Materials, Method E8, using a round tensile specimen with a two (2) inch gage length. The contractor shall submit a copy of all test reports monthly in accordance with Data Item Description DI-T-1906.

4.5.3 Slow bend precracked charpy test. Testing shall be conducted in accordance with the test procedure defined in the Appendix of this specification. The contractor shall submit a copy of all test reports monthly in accordance with Data Item Description DI-T-1906 reporting all information required by the Appendix A.

4.5.4 Steel quality. The Macroetch test shall be performed in accordance with supplementary requirement S10 of ASTM-A-711.

4.5.5 Radius of base of cavity. The warheads used in 4.4.3.1.2 shall be gaged for proper radius when the warheads are cut for tension specimens. If any warhead fails to comply, the heat treated inspection lot shall be rejected.

4.5.6 Interior surface. The warheads used in 4.4.3.1.2 shall be visually examined for interior surface discontinuities when the warheads are cut for tension specimens. Failure to meet the requirements of 3.5 shall be cause for rejection of the heat treated inspection lot.

4.5.7 Hydrostatic test. The unpainted warhead shall be mounted in a suitable test fixture in accordance with Dwg. 9287968 and be subjected to the pressure specified on the applicable requirements shall be classed defective and removed from the lot.

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4.5.8 Ultrasonic inspection. The warheads shall be ultrasonically inspected prior to threading using equipment and procedures in accordance with 4.5.8.1. The warhead shall be inspected with ultrasonics throughout all of the material. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. Two standards in accordance with 4.5.8.3 shall be used to check the effectiveness of the inspection process. Any warhead failing to meet the requirements of 3.7 shall be classed defective and removed from the lot.

4.5.8.1 Equipment and procedures. The ultrasonic inspection shall be performed with the use of a liquid to assure a reliable coupling to transmit ultrasonic energy. Should a stream or thin film of liquid be used to couple this energy, controls shall be used to monitor the coupling and assure its effectiveness. Such monitoring is not required if submergence in a liquid is used.

Using ultrasonic energy at right angles requires at least two separate sources of ultrasonic energy. This means that in the ogive and bourrelet regions, ultrasonic energy shall move within the walls of the warhead with separate significant axial and circumferential components of velocity to detect flaws located anywhere on the outside, inside or within the walls. For the base region, significant ultrasonic velocity components shall separately be radial and circumferential.

Criteria for detection of cracks are grooves which simulate actual larger cracks. Details are contained on Drawings 9246488 and 9246489. Criteria for detection of piping in the base region consists of three drilled holes oriented parallel to the major axis of the warhead. During warhead rotation, the hole that comes second closest to the transducer is the target for locating the ultrasonic focal zone.

All grooves and holes machined on the standards are targets for rejection. That is, any warhead that sends back an echo signal that is greater than or equal to the signal from the standard shall be rejected. Relief portions of radially oriented grooves where the full depth approaches the surface are not inspection criteria.

4.5.8.2 Calibration and operation procedures and equipment. Prior to initiation of ultrasonic inspection, the contractor shall submit for written approval one copy of his proposed procedures, including descriptions of equipment (contractor design) and operation procedures (see 6.3). No change to the

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procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum, the adequacy of the inspection process shall be checked by using the standards at the start and finish of each production shift or change in operator, as well as at four (4) hour intervals during production. All channels shall properly indicate a rejection signal when encountering the notches and holes mentioned in 3.7. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected. Equipment used shall provide for a positive activation of a buzzer or alarm as well as a light system for each channel to alert the operator of the signal indications that are the criteria for rejection. A log book shall be maintained by the operator of each system which is a minimum, documents the use of standards, transducers, instrumentation, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

4.5.8.3 Ultrasonic standards. Ultrasonic inspection standards shall be provided by the contractor for use during the ultrasonic inspection. These standards shall be machined from acceptable warheads in accordance with Drawings 9246488 and 9246489 and shall be inspected for all drawing requirements to determine their acceptability for use. Government verification of all inspections performed on the standards is required. In addition, a certificate of conformance for the standards shall be submitted to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801, prior to use of the standards.

4.5.8.4 Operator qualification. Personnel performing the ultrasonic inspection shall be certified in accordance with MIL-STD-410 with the following exception:

Each facility shall have as a minimum an individual who qualified as a Level II inspector in all respects but need not include experience.

4.5.9 Visual examination. The internal cavity of the warhead shall be inspected visually using peripheral lighting to assure conformance with the applicable requirements. Visual standards in accordance with 3.5.3 shall be set up for use during this inspection. Any warhead failing to meet the applicable requirements of 3.5 shall be classed defective and removed from the lot.

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4.5.10 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141 and TT-C-490 with the exception that test samples shall be dried in accordance with Drawing 9287943.

4.5.11 Protective coating - thickness. Protective coating thickness shall be accomplished using commercially available equipment.

4.5.12 Bonding assembly pull test fixture. A fixture using eitner suction or vacuum cups capable of exerting a minimum of 100 pounds pull on the insulator shall be used for the test. The fixture is to be designed in accordance with 4.4.4.

4.5.13 Proving ground. These tests shall be performed at a Government proving ground in accordance with the applicable acceptence test procedure. The sample warheads selected in accordance with 4.4.3.8 shall be inert loaded and assembled and fired at ambient temperature in a M201 series cannon with muzzle brake using a propelling charge adjusted which produces an average pressure of 47,000 + 1000, -1500 PSI chamber pressure based on copper pressure gages. The propelling charges shall be uniformly temperature conditioned prior to the test. Pressure-time traces and differential pressures shall be obtained for information only for each round fired by means of a spindle tap in the breech of the photographic and instrumental coverage shall be provided to assure compliance with 3.13. Range and deflections shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts for informational purposes only.

5. PREPARATION FOR DELIVERY.

5.1 Packing. Level C - The Aft Warhead Section and Strippable Coating assembly shall be packed in accordance with Dwg. 9280132.

5.2 Marking. Marking shall be in accordance with Dwg. 9280132.

5.3 Data cards. See MIL-A-48078 (PA).

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6. NOTES.

6.1 Intended use. The components covered by this specification are intended for use on the 8-Inch, M650, RA Projectile.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures and/or equipment designs for approval. See MIL-A-48078. Submit equipment designs, as requested, to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured to gaged to determine acceptability.

6.6 Proving ground test summary.

Test	Sample Size (see 4.4.3.8)		Requirements
Security	*20	10	No break-up or metal parts separation (see 3.5).

* Applicable only for First Article Test and first three lots.

6.7 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition cards shall be forwarded to Commander, ARRADCOM, ATTN: DRDAR-QAR-Q and DRDAR-LCU-CA, Dover, New Jersey 07801.

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6.8 Groove profile. Dow Corning silicone rubber RTV93-072 has been successfully used to obtain accurate profiles of grooves for inspection. The rubber was mixed, poured in such a way as to eliminate trapped air, cured for a day and then carefully removed. Very thin slices were taken across the impression of the groove near each end and in the center. Photos of projected image as seen on the screen of an optical comparator were used for confirmation.

6.9 Visual standards. Visual standards to be established at contractors plant in compliance with 3.5.3.

6.10 Forging cooling. Cooling of the forgings should not be retarded. Slow cooling can cause the formation of carbides.

6.11 Spherordizing. Spherordizing after forging can be beneficial in rough machining, nosing and heat treatment.

6.12 Quenching. Quenching over spuds is recommended to assure adequate quenching of the base.

6.13 Hydrostatic test plug. Examination should be made of a hydrostatic plug after failure of a warhead. A badly nicked or damaged plug may cause failure of an acceptable warhead.

6.14 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.5.2	Test Report	DI-T-1906
4.5.3	Test Report	DI-T-1906

(Copies of data item descriptions required by the contractors in conjunction with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.15 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Preparing Activity:
Army-AR

Project No.: 1320-A595

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APPENDIX

TEST METHOD FOR SLOW BEND PRECRACKED CHARPY TEST

10. SCOPE.

10.1 SCOPE. This appendix covers the test method for determining fracture toughness using a Charpy specimen containing a fatigue crack and tested in slow bending.

20. APPLICABLE DOCUMENTS.

20.1 ASTM Standards

E-4 Verification of Testing Machines

E-139 Recommend Practice for Conducting Creep, Creep-Rupture and Stress Rupture Tests of Metallic Materials.

E-399 Plane Strain Fracture Toughness of Metallic Materials.

30. SUMMARY OF METHOD.

30.1 The method involves three point bend testing of notched and fatigue cracked specimens using fixtures which minimize the contribution of friction forces to the measured applied load. The load-deformation characteristics of the test is recorded and the fracture toughness is determined from these characteristics and the original dimensions of the specimen.

40. PECAUTIONARY NOTES.

40.1 PRECAUTIONARY NOTES. When preparing specimens and testing, it is important that certain precautions be kept in mind.

40.1.1 Crack strength. The crack strength of some materials will be sensitive to the rate of loading, and loading rates should be confined to the range given in 80.3.2.

40.1.2 Temperature. The crack strength will be influenced by the testing temperature, and calibration relations may change with testing temperature.

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50. DEFINITIONS.

50.1 a - Fatigue crack depth, Inches (in)

B - specimen thickness (in)

W - specimen depth (in)

M - Bending Moment, Inches-Pound Force (in - lb)

f (a/W) - Power series of ration (a/w) from ASTM-E-399

S - Span between support rollers; (in)

P max - Maximum Load Applied, (lb)

Pp - A specific point at which the first load instability indicating unstable crack growth is recorded (lbs)

K - Stress intensity factor (ksi $\sqrt{in}$)

K_f (max) - Maximum stress intensity factor in fatiguing (ksi $\sqrt{in}$)

K_Q - Calculated fracture toughness (ksi $\sqrt{in}$)

K_{IC} - Plane strain fracture toughness (ksi $\sqrt{in}$)

60. APPARATUS.

60.1 Tension testing machine. Tension testing machine conforming to the requirements of ASTM-E-4.

60.2 Bend-Test fixture. The procedure involves the testing of the specimen in three-point bending. The bend fixture should be so designed as to minimize errors that arise from friction between the specimen and the supports. This can be accomplished by allowing the support rollers to rotate and move apart slightly during loading as described in Figure 1 of ASTM-E-399. The nominal span shall be 1.97 in. If the diameter of the support rollers is between 0.30 in. to 0.40 in., there will be a negligible change in the span during deformation of the specimen.

60.3 Temperature control. When testing at -65°F, an integral temperature conditioning chamber shall be used. Any suitable method may be used to temperature condition the

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specimen providing the region including the notched and cracked section can be uniformly maintained within ± 2°F of the desired test temperature. The cooling medium shall be free of contaminants, such as chloride or water, which have an adverse effect on the staballoy. When testing 70°F, no chamber is necessary, but the temperature must be held to plus or minus 8°F and humidity maintained below 40% RH.

60.4 Temperature measurement. The device used to measure the temperature of the medium should be placed adjacent to the test specimens. If an agitated liquid medium is used, the specimens shall be held at the desired temperature, plus or minus 2°F, for at least 5 minutes. Specimens being tested in gas shall be held at the desired temperature, plus or minus 2°F, for at least 60 minutes. See ASTM-E-139 for a discussion of errors on measurements.

60.4.1 Calibration. The temperature measuring apparatus should be calibrated periodically against standards traceable to the National Bureau of Standards in order to ensure that accuracy specified above can be achieved.

70. TEST SPECIMENS, DIMENSIONS AND PREPARATION.

70.1 Dimensions. The precracked Charpy bend specimen is shown in Figure 1. The crack depth (including notch) (a) is nominally equal to one-half the thickness (b) and shall lie within the range between 0.45 to 0.55 B.

70.1.1 Crack starter. The crack starter may consist of a "V" notch having an included angle between 45° and 60° provided with a small radius at its tip. Experience has shown that a notch radius of less than 0.003 in. and minimum machining surface compression stress is necessary in order to facilitate the initiation of a fatigue crack at a low stress intensity level.

70.2 Fatigue cracking. In order to ensure a sufficiently sharp fatigue crack and to facilitate the production of this crack, the following procedure shall be followed.

70.2.1 Specimen condition. The fatigue cracking shall be conducted with the specimen in the condition of heat treatment in which it is to be tested.

70.2.2 Cracked method. The method used for fatigue cracking shall be such that the crack extension direction is perpendicular to the notched face of the specimen. Cantilever

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bending or three point bending may be used to produce fatigue cracks. Cantilever bending facilitates completely reversed loading which aids in starting and propagating the crack. For practical purposes, the K calibration for three point bending may be used to estimate the stress intensity factor.

The expression given in ASTM-E-399 for the K calibration in three point bending may be rewritten in terms of the bending moment M as follows and used to compute the stress intensity in cantilever bending: $K = (4M/BW^{3/2}) f(a/W)$ where the values of the function f (a/w) are given in tabular form in ASTM-E-399.

The stress distribution at the crack tip in cantilever bending can be influenced by excessive clamping forces to the extent that the fatigue crack will deviate from a plane normal to the notched face of the specimen.

70.2.3 Cracking. During the last 0.020 in. of fatigue cracking, the maximum intensity of the fatigue cycle, K_f (max) shall not exceed 15.0 Ksi $\sqrt{\text{in.}}$ The stress intensity range should not be less than $0.9K_f$(max).

80. TEST PROCEDURE.

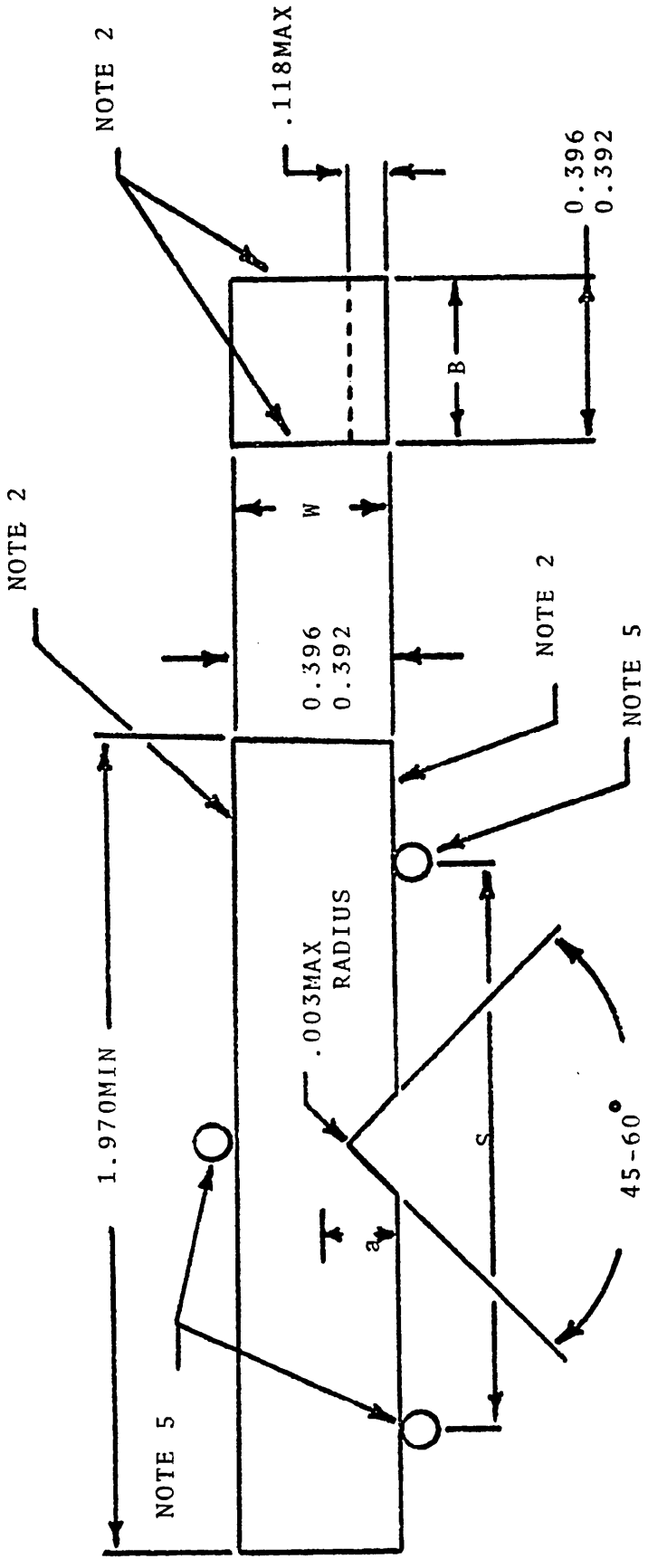
80.1 Number of tests. Samples shall be selected as specified in the technical data package.

80.2 Specimen measurement. All specimen dimensions shall be within the tolerances shown in Figure 1.

80.2.1 Measure the thickness (B) to the nearest 0.001 in. at not less than two positions between the fatigue crack tip and the unnotched edge of the specimen and record the average positions, one on each side of the crack starter notch and record the average value.

80.2.2 After fracture, measure the initial fatigue crack depth (a) to the nearest 0.5 percent at the following three positions: at the center of the crack front and midway between the center and the inter-section of the crack front with the specimen surfaces. Use the average of these three measurements as the crack depth in the calculation of K_Q (see 90.1). If the difference between any two of the crack length measurements exceeds 5 percent of the average, or if the part of the crack front is closer to the machined notch root than 5 percent of the average, the specimen should be discarded. Also, if the length of either surface trace of the crack is less than 80 percent of the average crack length, as defined above, the specimen should be discarded.

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NOTE1. All dimensions in inches.
NOTE2. Surfaces shall be perpendicular or parallel as applicable to within 0.001 in TIR.
NOTE3. Crack starter shall be perpendicular to specimen length and thickness to within ±2DEG.
NOTE4. Finish all over 63.
NOTE5. Bend fixture support rollers.

PRECRACKED CHARPY SPECIMEN SCALE

FIGURE 1

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80.3 Bend testing. Set up the bend test fixture so that the line of action of the applied load shall pass midway between the support role centers within 0.025 in. of the distance between these centers. Measure the span (distance between the support role centers) within 1 percent and record this value as S. This value should all between 1.49 in. and 1.65 in. If not, the span should be adjusted to fall within this range.

80.3.1 Locate the specimen with the crack tip midway between the roll centers to within 0.025 in. and square to the roll axis within 2 degrees. Check the parallellism between the back face of the specimen and the loading roller by sighting the gap between these surfaces just before contact is made. As a small load is applied, light in this gap should extinguish simultaneously across the thickness of the specimen. If this condition is not met, the loading roller should be suitably adjusted.

80.3.2 Load the specimen at a rate between 700 lb_f/min and 3600 lb_f/min until the maximum load is reached and autographically record the specimen load-displacement characteristics.

90. CALCULATION AND INTERPRETATION OF RESULTS.

90.1 Calculations: Calculate the fracture toughness of the specimen in compliance with the method and requirements outlined in ASTM-E-399. If a pop-in occurred, the minimum pop-in load P_p will be used as the basis for the calculation. For the bend specimen, calculate K_Q in units of psi-$in^{1/2}$ as follows:

$$K_Q = (P_p\ S/BW^{3/2})\ (2.9\ (a/W)^{1/2} - 4.6$$

$$(a/W^{3/2} + 21.8\ (a/W)^{5/2} - 37.6\ (a/W)^{7/2} + 38\ (a/W)^{9/2})$$

$$K_Q = P_p\ S/BW^{3/2}\ f\ (a/W)$$

The power series f(a/W) are tabulated in the following table for specific values of a/W:

a/W	f(a/W)	a/W	f(a/W)
0.450	2.28	0.500	2.66
0.455	2.32	0.505	2.71
0.460	2.35	0.510	2.75
0.465	2.39	0.515	2.84
0.475	2.46	0.525	2.89
0.480	2.50	0.530	2.94
0.485	2.54	0.535	2.99
0.490	2.58	0.540	3.04
0.495	2.62	0.545	3.09
		0.550	3.15

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$K_Q = K_{IC}$, if the calculated value of 2.5 $(K_Q/ y.s)^2$ is less than both the thickness and the crack length of the specimen. Whenever the sample fails to meet this requirement, the contractor shall refer the results to the contracting officer.

90.2 Fracture appearance. The appearance of the fracture may furnish valuable supplementary information. A means for describing the fracture appearance of cracked bend specimens is given in ASTM-E-399 and may be used to characterize the fracture appearance of the precracked Charpy specimens.

100. REPORT.

100.1 The report shall include the following for each specimen tested.

100.1.1 All specimen dimensions measured.

100.1.2 Span between support rollers.

100.1.3 The maximum load P_{max} and the minimum pop-in load P_p, if applicable.

100.1.4 Fracture toughness ($K_Q = K_{1c}$).

100.1.5 Test temperature.

100.1.6 The maximum stress intensity $K_{f\ (max)}$ during the final 0.020 in. of fatigue cracking.

100.1.7 Crack plane orientation.

100.1.8 Fracture appearance.

100.1.9 0.2 percent offset tensile-yield strength.

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